Microsoft Cloud Forensic Evidence Acquisition

PowerShell Reference for Microsoft Cloud and Endpoint Evidence

How can an investigator acquire or query each Microsoft evidence source with PowerShell?

11 evidence sources · 8 Microsoft products · 154 official Microsoft references
Checked against Microsoft Learn on 2026-10-13

PowerShell acquisition cannot recreate telemetry that was never generated, enabled, collected, retained, or made accessible to the investigator.

No evidence does not automatically mean no activity occurred.

COMMAND TAGS

ACQUIRE	Acquires evidence
INSPECT	Inspects configuration
CHANGES CONFIG	Changes configuration
CONNECT	Authenticates session

Interface status follows each command name: GA, Current, Beta, Preview, Retiring. Beta and preview interfaces can change without notice.

VERIFICATION STATUS

✓ Verified	Every field checked against current Microsoft documentation
! Conditional	Usable, with listed gaps still to confirm
? Verification required	Not yet confirmed; do not rely on it
✕ Deprecated or replaced	Retiring or replaced; use the named successor

"Verification required" marks a value Microsoft documentation did not confirm. Test it before relying on it.

AVAILABILITY LEGEND

AUTO	Evidence is generated or collected automatically
CONFIG	Auditing, collection, telemetry, or a feature must be configured
EXPORT	Native evidence can be exported or routed
LICENSE	Availability depends on licensing
DEPLOY	Product deployment or onboarding is required
POLICY	Evidence detail depends on an audit or logging policy
ON DEMAND	Evidence is produced only when requested
SUPPORTING	Contextual evidence rather than a primary audit trail

1 Native PowerShell cmdlets

Az PowerShell, Defender, and documented product commands that read evidence or its configuration directly.

02 AZURE MONITOR · ! Conditional · 1 open gap

Azure Activity Log

ACQUISITION METHOD Az PowerShell
API Activity Log REST API (Azure Resource Manager) (GA)
POWERSHELL MODULE OR INTERFACE `Az.Monitor`, `Az PowerShell (Activity Log REST API)`

CMDLET, API, OR QUERY MECHANISM

ACQUIRE **Get-AzActivityLog** Current

```
Get-AzActivityLog -StartTime ([datetime]"2026-10-01T00:00:00Z") -EndTime
([datetime]"2026-10-07T23:59:59Z") | ConvertTo-Json -Depth 10 | Out-File activity-
log.json
```

Paging and limits: Microsoft's examples return at most 1,000 events per call; narrow the time range or set -MaxRecord. Both ends of the range must fall inside the 90-day retention window.
Parameter sets scope by subscription (default), -ResourceGroupName, -ResourceId, -ResourceProvider, or -CorrelationId; -Caller and -Status filter further; -DetailedOutput returns full properties. Microsoft's examples export with Export-Csv; JSON keeps nested claims and properties intact.

ACQUIRE **Invoke-AzRestMethod** GA

```
Invoke-AzRestMethod -Method GET -Path
"/providers/Microsoft.Insights/eventtypes/management/values?api-version=2015-04-
01&`$filter=eventTimestamp ge '2026-10-01T00:00:00Z' and eventTimestamp le '2026-10-
07T23:59:59Z'"
```

Paging and limits: Verification required
Tenant-level events are separate from subscription events and can include management group or subscription creation.

ACQUIRE **Invoke-AzRestMethod** Preview

```
Invoke-AzRestMethod -Method GET -Path
"/providers/Microsoft.Management/managementGroups/<managementGroupId>/providers/Microsoft
.Insights/eventtypes/management/values?api-version=2017-03-01-
preview&`$filter=eventTimestamp ge '2026-10-01T00:00:00Z' and eventTimestamp le '2026-10-
07T23:59:59Z'"
```

Paging and limits: Verification required
Microsoft documents the 2017-03-01-preview API version as required for management-group queries.

REQUIRED PERMISSIONS
- **Azure RBAC:** Monitoring Reader (can query the Activity Log). A custom role with Microsoft.Insights/eventtypes/* limits access to Activity Log reads.

PRIOR-COLLECTION REQUIREMENT
AUTO EXPORT **Automatic with limited native retention**
Collected by default with no configuration.
Native retention: 90 days

RECOMMENDED OUTPUT FORMAT JSON CSV

IMPORTANT LIMITATIONS
- Retained for 90 days natively; both ends of a query time range must fall inside that window.
- Does not typically capture read operations. Operations inside a resource are in Resource Logs.
- Tenant- and management-group-level scopes only capture Azure Resource Manager events in those hierarchies.
- Values in AzureActivity fields can differ in case; use case-insensitive operators.
- Entries are usually available 3 to 20 minutes after the event.
- The Activity Log REST API is limited to 50 queries per 30 seconds with a 75-second maximum timeout; querying AzureActivity in a workspace counts against the general Logs query limits instead.

04 NETWORK WATCHER · ! Conditional · 2 open gaps

Virtual network flow logs

ACQUISITION METHOD Native PowerShell cmdlets (additional route)
POWERSHELL MODULE OR INTERFACE `Az.Storage`

CMDLET, API, OR QUERY MECHANISM

ACQUIRE **Get-AzStorageBlobContent** Current

```
Get-AzStorageBlob -Container 'insights-logs-flowlogflowevent' -Context <storage-context>
| Get-AzStorageBlobContent -Destination C:\Evidence\FlowLogs
```

Paging and limits: Get-AzStorageBlob lists the blobs; Microsoft recommends -MaxCount when a container holds many blobs.
Blob path: insights-logs-flowlogflowevent/flowLogResourceID=/{subscriptionID}_NETWORKWATCHERRG/NETWORKWATCHER_{Region}_{ResourceName}-{ResourceGroupName}-FLOWLOGS/y=/m=/d=/h=/m=00/macAddress=/PT1H.json. Hash each downloaded file.

Permissions, prior-collection requirement, output, and limitations: see this source under path 3.

07 DEFENDER ANTIVIRUS · ✓ Verified

Microsoft-Windows-Windows Defender/Operational event log

ACQUISITION METHOD Native PowerShell cmdlets (additional route)
POWERSHELL MODULE OR INTERFACE `Defender`

CMDLET, API, OR QUERY MECHANISM

ACQUIRE **Get-MpThreatDetection** Current

```
Get-MpThreatDetection
```

INSPECT **Get-MpComputerStatus** Current

```
Get-MpComputerStatus | Select-Object AMRunningMode
```

Permissions, prior-collection requirement, output, and limitations: see this source under path 4 on Sheet B.

14 AZURE MONITOR · ! Conditional · 1 open gap

Diagnostic Settings

ACQUISITION METHOD Az PowerShell
API Azure Resource Manager (Microsoft.Insights/diagnosticSettings) (Current)
POWERSHELL MODULE OR INTERFACE `Az.Monitor`

CMDLET, API, OR QUERY MECHANISM

INSPECT **Get-AzDiagnosticSetting** Current

```
Get-AzDiagnosticSetting -ResourceId <resourceId> | ConvertTo-Json -Depth 10
```

Returns the active (current) settings, not their history.

INSPECT **Get-AzDiagnosticSettingCategory** Current

```
Get-AzDiagnosticSettingCategory -ResourceId <resourceId>
```

CHANGES CONFIG **New-AzDiagnosticSetting** Current

```
New-AzDiagnosticSetting -Name <name> -ResourceId <resourceId> -WorkspaceId
<workspaceResourceId> -Log $log -Metric $metric
```

Changes the environment. Record who ran it and when. It does not create logs for activity that happened before it existed.

REQUIRED PERMISSIONS
- **Azure RBAC:** Monitoring Reader to view diagnostic settings; Monitoring Contributor to create or edit them, plus ListKeys on a storage account or Event Hubs destination. Read access to the resource type and scope is also needed.
- **Microsoft Entra role:** Security Administrator (least privileged to configure Microsoft Entra diagnostic settings)

PRIOR-COLLECTION REQUIREMENT
CONFIG **Configuration required**
A diagnostic setting names the categories (or a category group: allLogs, or audit for logs that record interactions with data or service settings) and up to one destination of each type: Log Analytics workspace, Azure Storage account, Event Hubs, or a partner solution.
Native retention: Not applicable (configuration object)

RECOMMENDED OUTPUT FORMAT JSON

IMPORTANT LIMITATIONS
- Shows current configuration only; on its own it does not prove what was configured at the time of an incident.
- Some services need more than one diagnostic setting to collect both legacy and resource-specific logs (for example, Azure Firewall).
- Switching an existing setting to resource-specific mode leaves earlier data in AzureDiagnostics.
- Data starts flowing up to 90 minutes after a setting is created.
- If a resource is deleted or renamed and later re-created with the same name, a leftover diagnostic setting can apply to the new resource and resume collection. Microsoft advises deleting diagnostic settings with the resource.
- Diagnostic settings don't allow filtering within a category; transformations filter supported tables in the workspace.

14 SENTINEL · ! Conditional · 1 open gap

Microsoft Sentinel analytics, alerts, and incidents

ACQUISITION METHOD Native PowerShell cmdlets (additional route)
POWERSHELL MODULE OR INTERFACE `Az.SecurityInsights`

CMDLET, API, OR QUERY MECHANISM

ACQUIRE **Get-AzSentinelIncident** Current

```
Get-AzSentinelIncident -ResourceGroupName <resource-group> -WorkspaceName <workspace-
name>
```

Paging and limits: Verification required

ACQUIRE **Get-AzSentinelIncidentAlert** Current

```
Get-AzSentinelIncidentAlert -ResourceGroupName <resource-group> -WorkspaceName
<workspace-name> -IncidentId <incident-id>
```

Permissions, prior-collection requirement, output, and limitations: see this source under path 5 on Sheet B.

3 Azure Monitor and KQL

Evidence already centralized in a Log Analytics workspace or Microsoft Sentinel. Check whether the source writes to AzureDiagnostics or a resource-specific table.

SHARED BY SEVERAL SOURCES IN THIS PATH

Required permissions
- **Azure RBAC:** Monitoring Reader (can search Log Analytics workspaces and run log queries)

Paging and limits
- No paging; Azure Monitor log query limits apply.
- No paging. Logs query API limits: 500,000 records and about 100 MB per query, 10-minute maximum run time (3-minute default), 200 requests per 30 seconds per user. Split large pulls by time range.

Important limitations
- Monitoring Reader and Monitoring Contributor don't grant read access to log data in storage accounts or Event Hubs; reading an archived copy needs separate data access.

03 AZURE MONITOR · ✓ Verified

Azure Resource Logs

ACQUISITION METHOD Azure Monitor and KQL
API Azure Monitor Logs query API (Current)
POWERSHELL MODULE OR INTERFACE `Az.OperationalInsights`

CMDLET, API, OR QUERY MECHANISM

ACQUIRE **Invoke-AzOperationalInsightsQuery** Current

```
Invoke-AzOperationalInsightsQuery -WorkspaceId <workspace-id> -Query "<resource-specific
table> | union AzureDiagnostics" -Timespan (New-TimeSpan -Days 7)
```

Paging and limits: see the shared box for this path.
Replace the placeholder with the service's resource-specific table from the supported categories reference.

REQUIRED PERMISSIONS
- Plus the shared permissions for this path

PRIOR-COLLECTION REQUIREMENT
CONFIG EXPORT **Configuration required**
Create a diagnostic setting for each resource and select the categories to collect.
Native retention: None: not collected or stored by Azure Monitor unless a diagnostic setting routes them

RECOMMENDED OUTPUT FORMAT JSON CSV

IMPORTANT LIMITATIONS
- Not completely lossless: store-and-forward with retries but no transactional guarantees, so small losses can occur.
- Ingestion latency varies by destination and service.
- AzureDiagnostics is not the universal table. Most services write to resource-specific tables, and some let you choose the mode.
- Storage blobs are named by the time records were received, so a blob can contain records outside the hour in its path.
- Data starts flowing up to 90 minutes after a diagnostic setting is created; nothing is collected for the period before it existed.
- Plus the shared limitations for this path

03 KEY VAULT · ! Conditional · 1 open gap

Key Vault resource logs (AuditEvent)

ACQUISITION METHOD Azure Monitor and KQL
API Azure Monitor Logs query API (Current)
POWERSHELL MODULE OR INTERFACE `Az.OperationalInsights`

CMDLET, API, OR QUERY MECHANISM

ACQUIRE **Invoke-AzOperationalInsightsQuery** Current

```
Invoke-AzOperationalInsightsQuery -WorkspaceId <workspace-id> -Query "AZKVAuditLogs" -
Timespan (New-TimeSpan -Days 7)
```

Paging and limits: see the shared box for this path.

REQUIRED PERMISSIONS
- Plus the shared permissions for this path

PRIOR-COLLECTION REQUIREMENT
CONFIG EXPORT **Configuration required**
Enable a diagnostic setting on the vault with the AuditEvent category (in the portal, category groups 'audit' and 'allLogs').
Native retention: None: not collected or stored until a diagnostic setting routes them

RECOMMENDED OUTPUT FORMAT JSON CSV

IMPORTANT LIMITATIONS
- Key Vault documentation calls AuditEvent the only Key Vault log category, while the Log Analytics table reference also lists a policy-evaluation table. Confirm the categories with Get-AzDiagnosticSettingCategory.
- Key Vault documentation examples use Set-AzDiagnosticSetting, while the current Az.Monitor reference documents New-AzDiagnosticSetting. Check against the installed Az.Monitor version.
- Key Vault appears in the AzureDiagnostics-mode service list and also has resource-specific tables; check both.
- Plus the shared limitations for this path

03 STORAGE · ✓ Verified

Storage resource logs (StorageRead, StorageWrite, StorageDelete)

ACQUISITION METHOD Azure Monitor and KQL
API Azure Monitor Logs query API (Current)
POWERSHELL MODULE OR INTERFACE `Az.OperationalInsights`

CMDLET, API, OR QUERY MECHANISM

ACQUIRE **Invoke-AzOperationalInsightsQuery** Current

```
Invoke-AzOperationalInsightsQuery -WorkspaceId <workspace-id> -Query "StorageBlobLogs" -
Timespan (New-TimeSpan -Days 7)
```

Paging and limits: see the shared box for this path.

REQUIRED PERMISSIONS
- Plus the shared permissions for this path

PRIOR-COLLECTION REQUIREMENT
CONFIG EXPORT **Configuration required**
Configure diagnostic settings on the account and on each data service (blob, file, queue, table) to be covered.
Native retention: None: not collected or stored until a diagnostic setting routes them

RECOMMENDED OUTPUT FORMAT JSON CSV

IMPORTANT LIMITATIONS
- Requests are logged on a best-effort basis.
- Of failed anonymous requests, only server errors, time-outs, and 304 (Not Modified) GET failures are logged.
- Requests made by the Blob service itself, such as log creation or deletion, aren't logged. Azure Monitor filters out activity in 'insights-logs-' containers.
- Operations the Azure portal performs while someone views the account are also logged.
- Plus the shared limitations for this path

04 AZURE FIREWALL · ✓ Verified

Azure Firewall resource logs

ACQUISITION METHOD Azure Monitor and KQL
API Azure Monitor Logs query API (Current)
POWERSHELL MODULE OR INTERFACE `Az.OperationalInsights`

CMDLET, API, OR QUERY MECHANISM

ACQUIRE **Invoke-AzOperationalInsightsQuery** Current

```
Invoke-AzOperationalInsightsQuery -WorkspaceId <workspace-id> -Query "AZFWNetworkRule |
union AZFWApplicationRule, AZFWNatRule, AZFWThreatIntel, AZFWIdpsSignature" -Timespan
(New-TimeSpan -Days 7)
```

Paging and limits: see the shared box for this path.

REQUIRED PERMISSIONS
- Plus the shared permissions for this path

PRIOR-COLLECTION REQUIREMENT
CONFIG EXPORT **Configuration required**
Requires a diagnostic setting.
Native retention: None: not collected or stored until a diagnostic setting routes them

RECOMMENDED OUTPUT FORMAT JSON CSV

IMPORTANT LIMITATIONS
- Legacy AzureDiagnostics categories put details in a free-text properties message; resource-specific tables are structured.
- Firewall logs cover traffic that passes through the firewall only.
- Plus the shared limitations for this path

04 NETWORK WATCHER · ! Conditional · 2 open gaps

Virtual network flow logs

ACQUISITION METHOD Azure Monitor and KQL
POWERSHELL MODULE OR INTERFACE `Az.OperationalInsights`

CMDLET, API, OR QUERY MECHANISM

ACQUIRE **Invoke-AzOperationalInsightsQuery** Current

```
Invoke-AzOperationalInsightsQuery -WorkspaceId <workspace-id> -Query "NTANetAnalytics |
where SubType == 'FlowLog'" -Timespan (New-TimeSpan -Days 7)
```

Paging and limits: see the shared box for this path.
Use only SubType FlowLog; other values are internal.

REQUIRED PERMISSIONS
- **Azure RBAC:** Verification required: Network Watcher and storage account permissions (Microsoft documents a Network Watcher RBAC permissions article) **Verification required**

PRIOR-COLLECTION REQUIREMENT
CONFIG EXPORT **Configuration required**
Create a flow log on the virtual network, subnet, or network interface and choose a storage account.
Native retention: Set per flow log in days on Standard general-purpose v2 storage accounts; 0 keeps data until manually deleted

RECOMMENDED OUTPUT FORMAT JSON CSV

IMPORTANT LIMITATIONS
- The storage account must be in the same region as the virtual network and be Standard tier.
- Rotating customer-managed keys on the storage account stops flow logging until it is disabled and re-enabled.
- Don't modify flow-log blobs while they are being written; it can break writes for that hour's blob.
- Traffic can't be recorded at a private endpoint; it is captured at the source VM.
- Outbound VM flows to ExpressRoute aren't recorded when flow logging is enabled on the ExpressRoute gateway subnet; FastPath traffic bypasses the gateway.
- Azure Container Instances aren't supported.

05 WINDOWS · ✓ Verified

Security event log

ACQUISITION METHOD Azure Monitor and KQL (additional route)
POWERSHELL MODULE OR INTERFACE `Az.OperationalInsights`

CMDLET, API, OR QUERY MECHANISM

ACQUIRE **Invoke-AzOperationalInsightsQuery** Current

```
Invoke-AzOperationalInsightsQuery -WorkspaceId <workspace-id> -Query "SecurityEvent |
where EventID == 4625 | summarize count() by TargetAccount" -Timespan (New-TimeSpan -Days
7)
```

Paging and limits: see the shared box for this path.
Use the Event table instead when the Security log was collected by a plain Azure Monitor Agent data collection rule.

Permissions, prior-collection requirement, output, and limitations: see this source under path 4 on Sheet B.

14 AZURE MONITOR · ! Conditional · 1 open gap

Log Analytics workspace (Azure Monitor Logs)

ACQUISITION METHOD Azure Monitor and KQL
API Azure Monitor Logs query API (Current)
POWERSHELL MODULE OR INTERFACE `Az.OperationalInsights`

CMDLET, API, OR QUERY MECHANISM

INSPECT **Get-AzOperationalInsightsWorkspace** Current

```
Get-AzOperationalInsightsWorkspace | Select-Object CustomerId, Name
```

ACQUIRE **Invoke-AzOperationalInsightsQuery** Current

```
$r = Invoke-AzOperationalInsightsQuery -WorkspaceId <workspace-id> -Query $query -
Timespan (New-TimeSpan -Days 30) -Wait 300; $r.Results | ConvertTo-Json -Depth 10 | Out-
File results.json
```

Paging and limits: No paging. Logs query API limits: 500,000 records and about 100 MB per query, 10-minute maximum run time (3-minute default), 200 requests per 30 seconds per user. Split large pulls by time range. If Metadata status is Pending, the search hasn't finished and results come from the archive.
Increase -Wait if the request is cancelled by the 100-second HttpClient timeout.

REQUIRED PERMISSIONS
- Plus the shared permissions for this path

PRIOR-COLLECTION REQUIREMENT
CONFIG **Configuration required**
Data arrives only through configured collection: Azure and Microsoft Entra diagnostic settings, data collection rules, and Microsoft Sentinel connectors.
Native retention: Analytics tables: 30 days by default (some log tables 90 days); analytics retention up to 2 years; total retention up to 12 years

RECOMMENDED OUTPUT FORMAT JSON CSV

IMPORTANT LIMITATIONS
- Holds only what was routed to it, from the time routing began.
- Query API limits: 500,000 records and about 100 MB per query, 10-minute maximum run time, 200 requests per 30 seconds per user. Search jobs can return up to 100,000,000 records per result set.
- Basic and Auxiliary tables have query restrictions.

14 SENTINEL · ! Conditional · 1 open gap

Microsoft Sentinel analytics, alerts, and incidents

ACQUISITION METHOD Azure Monitor and KQL (additional route)
POWERSHELL MODULE OR INTERFACE `Az.OperationalInsights`

CMDLET, API, OR QUERY MECHANISM

ACQUIRE **Invoke-AzOperationalInsightsQuery** Current

```
Invoke-AzOperationalInsightsQuery -WorkspaceId <workspace-id> -Query $query
```

Paging and limits: see the shared box for this path.

Permissions, prior-collection requirement, output, and limitations: see this source under path 5 on Sheet B.

RECORD EVERY ACQUISITION

- Source system and scope: tenant, subscription, workspace, resource, or device
- Query or filter used, including the time range
- UTC acquisition start and end time
- Collecting account or application identity, and its permission context
- Tool, module, or API name and version
- Original output format and any export format
- How pagination and result limits were handled
- Errors, throttling, and retries
- Known completeness limitations
- Hash of every file produced (for example, SHA-256)

Preserve structured raw results before creating summaries. An export mechanism does not establish legal chain of custody by itself.

OUTPUT FORMATS

JSON	APIs and structured cloud results; keeps nested properties
CSV	Analyst-friendly tables; flatten only after keeping the raw JSON
EVTX	Windows event channels in their original form
CLIXML	Only when PowerShell object fidelity matters
TXT	Human-readable supporting output, never the sole evidence format

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